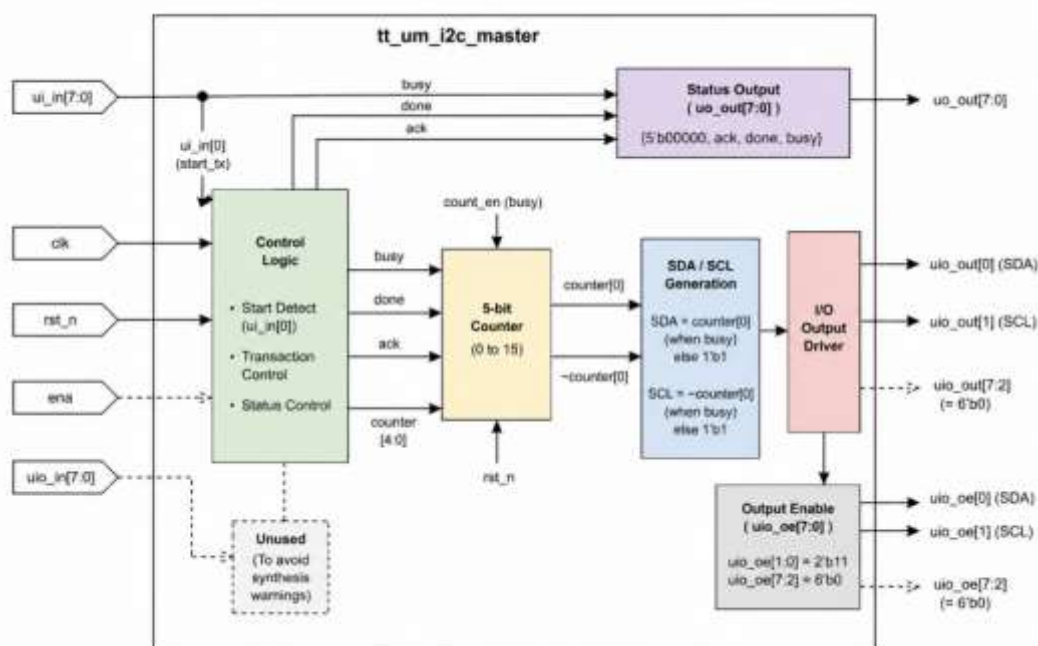


DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

Center of Excellence in VLSI Design

Simplified I²C Master RTL Architecture

The tt_um_i2c_master RTL is designed as a simple counter-based I²C transaction controller. An active-low reset initializes all internal registers (busy, done, ack, and counter). When the start signal (ui_in[0]) is asserted, the module begins a transaction by setting the busy flag and resetting the counter. During the transaction, the counter increments on each clock cycle and generates simulated SDA and SCL activity. Once the counter reaches 15, the transaction is completed, busy is cleared, and both done and ack flags are asserted. Status information is provided through the output bus, enabling external monitoring of the transaction state.



BLOCK DIAGRAM OF I2C MASTER



GDSII OF THE DESIGN I2C MASTER

THE TEAM AND THEIR CONTRIBUTIONS

A dedicated team of students has successfully developed a **Simplified I²C Master RTL Architecture**, demonstrating their growing expertise in digital integrated circuit design and semiconductor engineering. The project began with a detailed study of the I²C communication protocol, followed by RTL design using Verilog, functional verification through simulation, and iterative optimization to ensure reliable master-side communication with peripheral devices. To translate the design into a manufacturable integrated circuit, the students employed the **OpenLane** open-source ASIC design flow. The RTL design was synthesized into a gate-level netlist, followed by floorplanning, power planning, placement, clock tree synthesis, routing, design rule checking (DRC), layout-versus-schematic (LVS) verification, and timing analysis. This hands-on experience exposed the students to the complete RTL-to-GDSII design flow used in the semiconductor industry, bridging the gap between classroom concepts and real-world chip design. The project reflects the institution's commitment to developing industry-ready VLSI engineers with practical expertise in open-source Electronic Design Automation (EDA) tools and modern ASIC design methodologies.

The TEAM		
Mrs. Padmavathi N	Dr. Somashekhar G C	Mrs. Suchithra N P
Mr. Akash G G	Ms. Pushpalatha B	Ms. Dakshayini A V

CHIP TAPEOUT PROGRAM SCHEDULE

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HOW IS THIS PROJECT RELEVANT AND IMPORTANT FOR THE STUDENTS

The I²C protocol is one of the most widely used serial communication standards in embedded systems. Designing an I²C Master requires students to understand:

- Finite State Machine (FSM) design
- Clock generation and synchronization
- Serial communication protocols
- Bus arbitration concepts
- Timing and control logic
- RTL coding in Verilog

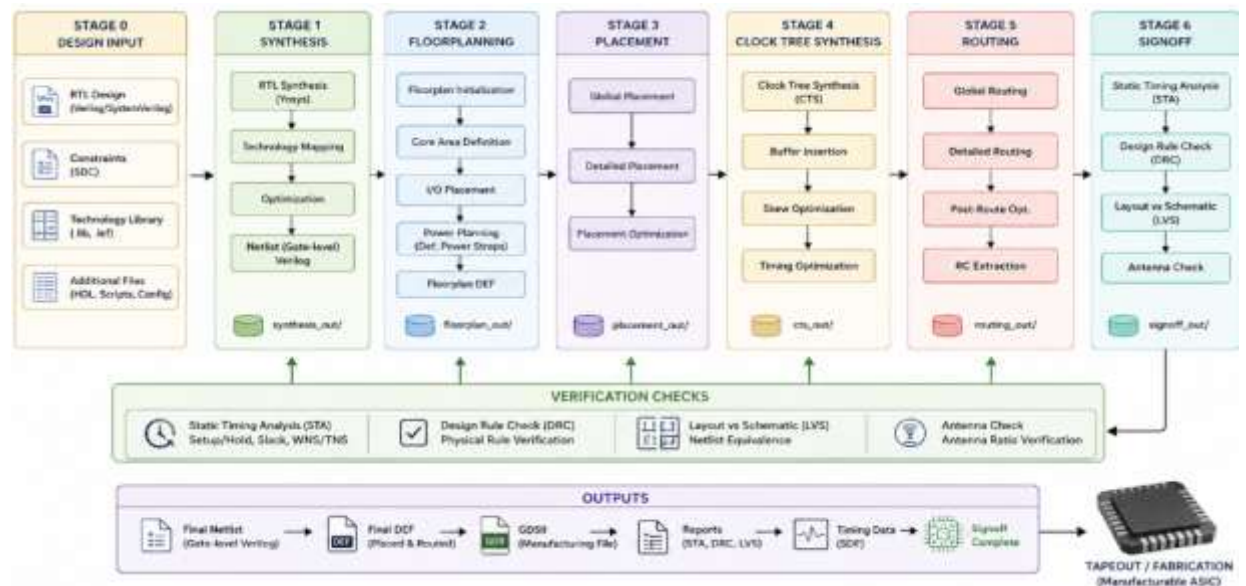
These are fundamental skills required for digital IC and FPGA design.

SCOPE FOR FUTURE ENHANCEMENT

The project also provides an excellent foundation for more advanced work. Students can extend it by:

- Supporting multi-master operation
- Adding clock stretching and arbitration
- Implementing I3C compatibility
- Integrating the controller into a RISC-V SoC
- Developing a UVM-based verification environment
- Comparing ASIC implementation with FPGA realization
- Optimizing for low power or high performance

THE ASIC COMPLETE FLOW



The Team Members



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Associate Prof



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Prof and HOD



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